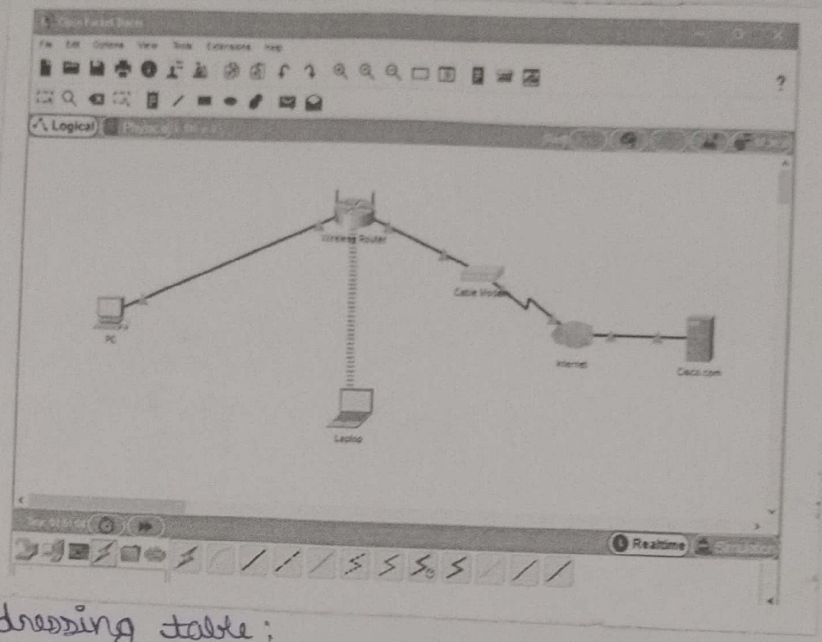
Ex.No:10b Design & Configure an internetwork using router, DHCP server.

Aim:

Design & Configure an internetwork using wireless router, DHCP server and internet cloud.



Addressing table:

Device	Interface	IP address	Subnet mask	default gateway
PC	ethernet 0.	DHCP		192,168.0.1
wireless router	Internet	DHCP		
laptop	wireless 0	DHCP.		

Objectives:

i) Build a simple network in the logical topology workspace.

ii) Configure the network devices.

iii) Test Connectivity between network devices.

iv) Save the file and close Packet Tracer.

i)

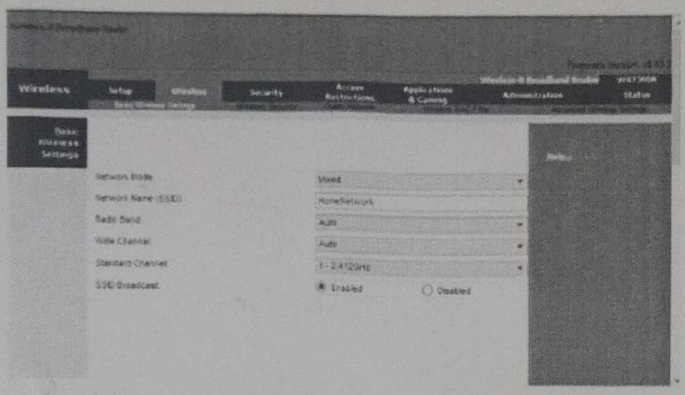
Step 1: Launch Packet Tracer.

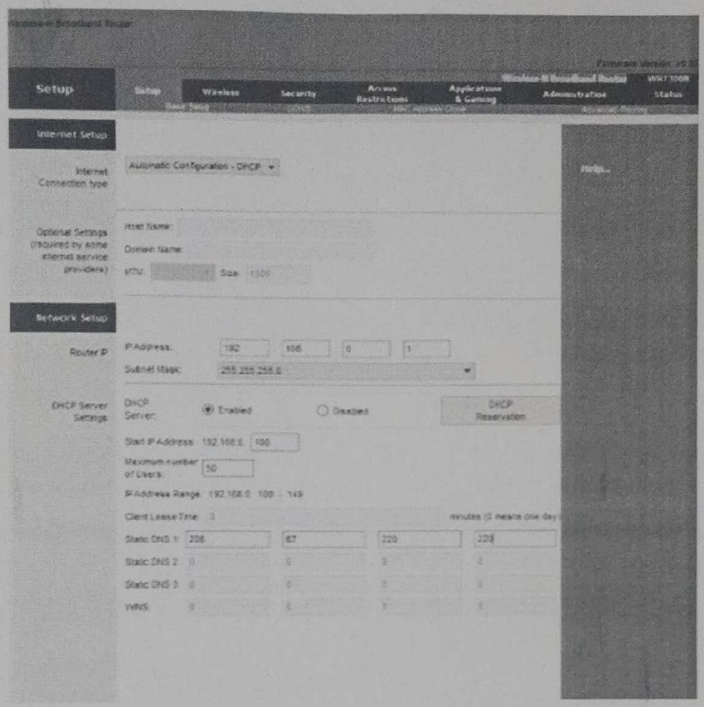
Step 2: Build the topology.

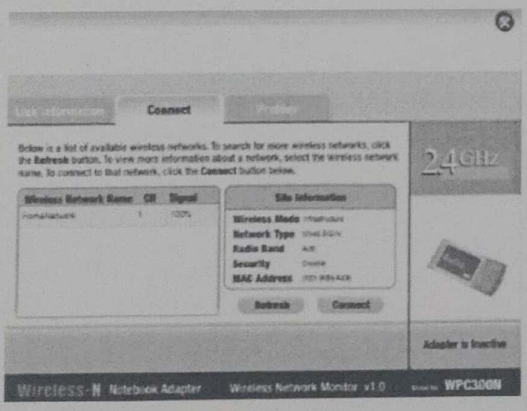
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ii) Step 1: Configure the wireless router,

Step 2: Configure the laptop.



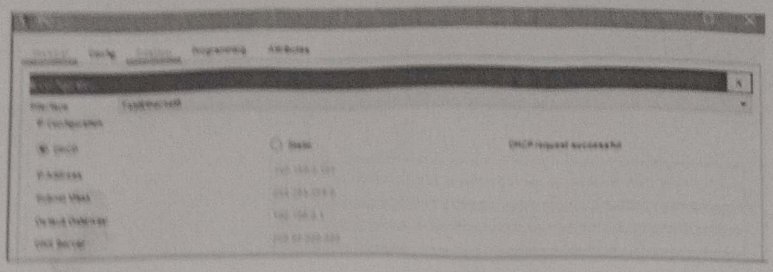




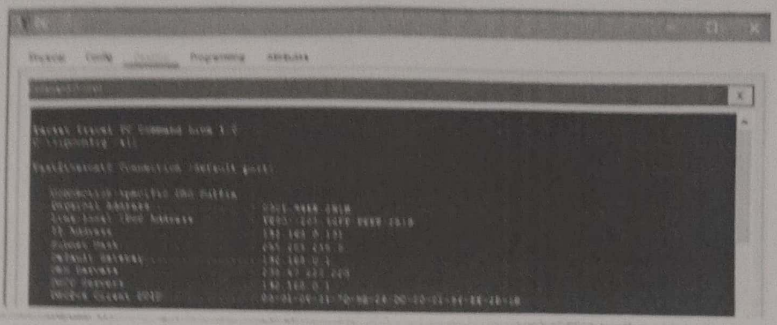
Step 3: Configure the PC,

Click on the PC icon on the packet tracer logical workspace and select the Desktop tab and then the IP configuration icon.

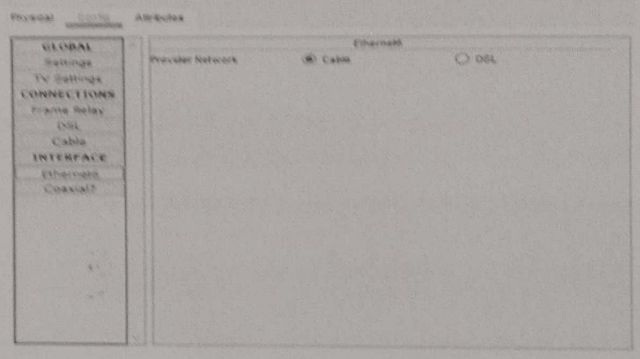
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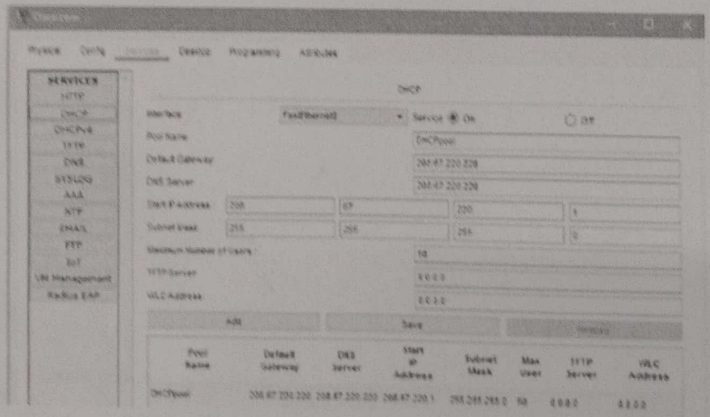


Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.



Step 4: Configure the Internet cloud.

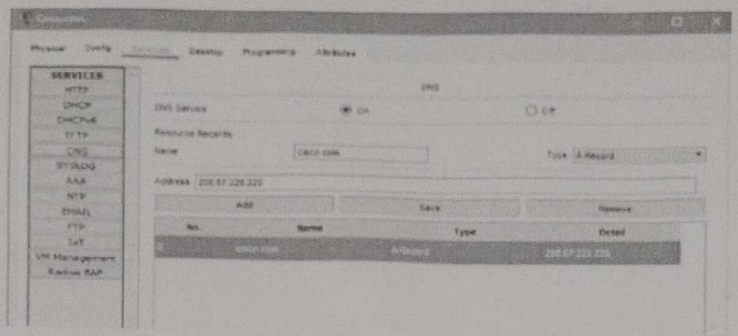


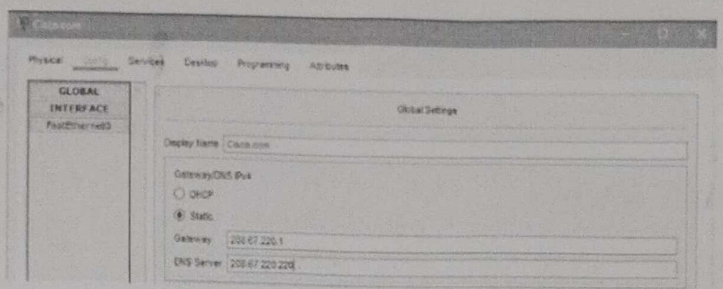


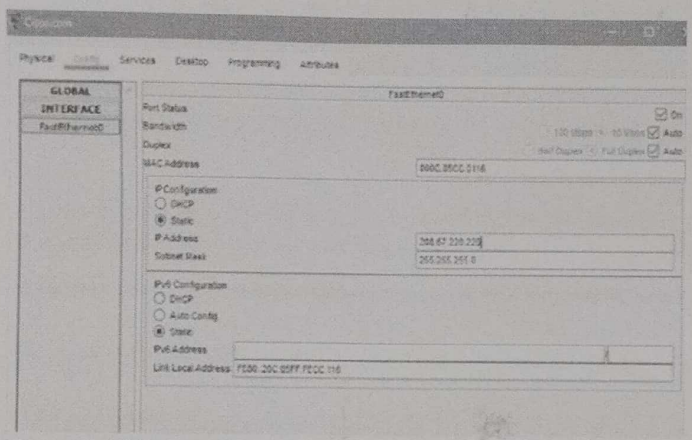
Step 5: Configure the Cisco.com server.

click add to add the pool.

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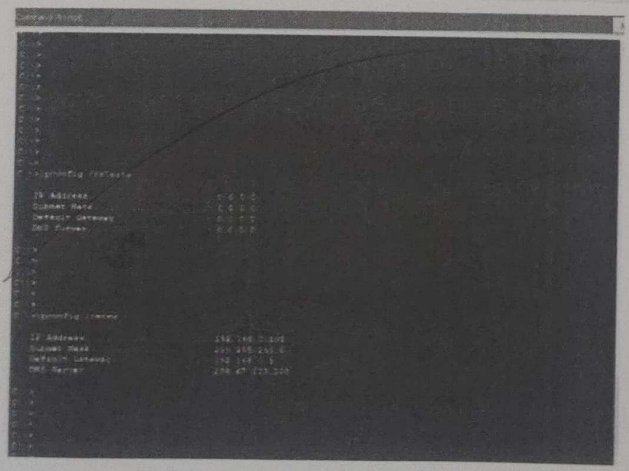




ii) Step 1:

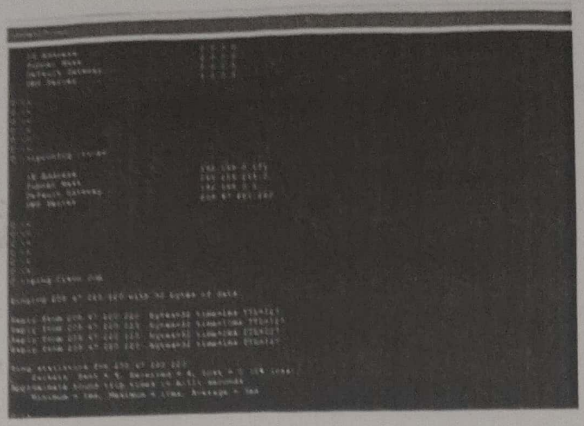
Refresh the IPV4 settings on the PC.

a) Verify that the PC is receiving IPV4 configuration information from DHCP.



b) Test connectivity to the Cisco.com server from the PC.

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Student Observation;

1) write down the key features of configuring wireless router and DHCP server.

⇒ Wireless router;

* Provides wireless network connectivity.
* Configures SSID.

DHCP server;

* Assigns DNS server addresses.
* Avoids IP address conflicts.

2) What is the significance of DHCP server in inter-networking?

⇒ * Automatically assigns IP addresses.
* Reduces manual configuration.
* Simplifies network administration.
* Prevents duplicate IP address conflicts.

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